LAB PROGRAM

CHAPTER 2

1. Write a program to perform the following

o An empty list

o A list with one element

o A list with all identical elements

o A list with negative

numbers Test Cases:

1. Input: []

o Expected Output: []

2. Input: [1]

o Expected Output: [1]

3. Input: [7, 7, 7, 7]

o Expected Output: [7, 7, 7, 7]

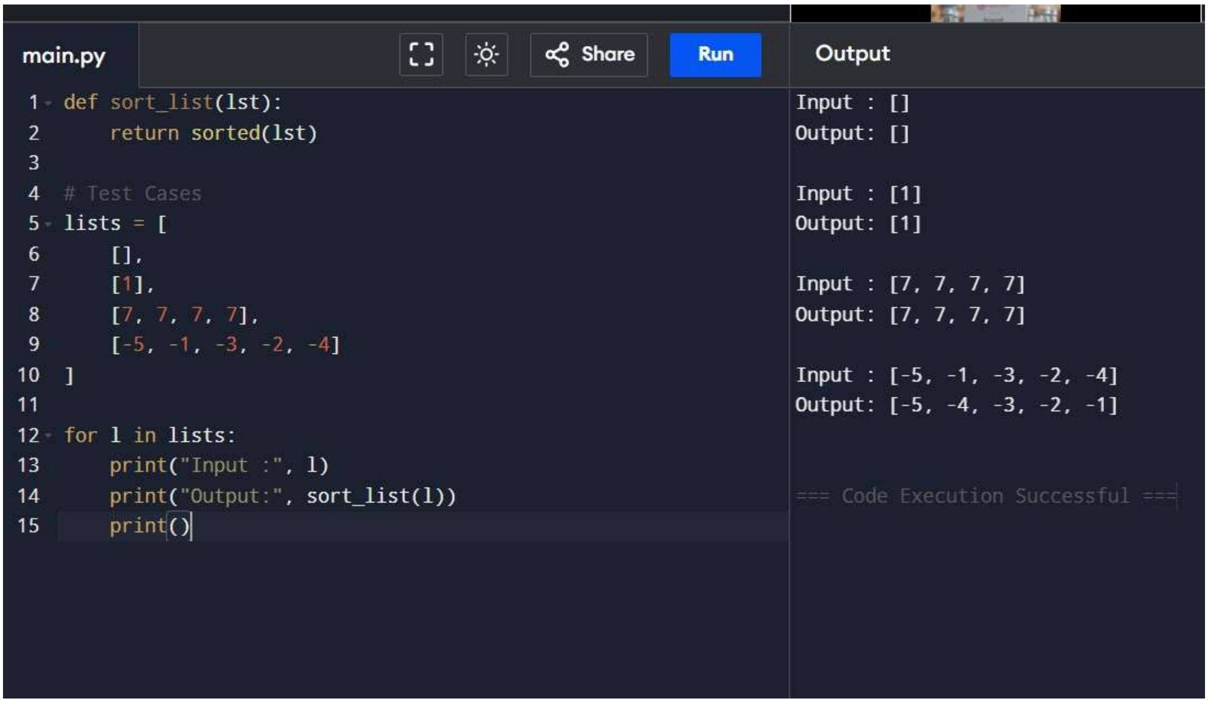
4. Input: [-5, -1, -3, -2, -4]

o Expected Output: [-5, -4, -3, -2, -1]

CODE:

```
def sort_list(lst):
    return
    sorted(lst)
lists = [
    [],
    [1],
    [7, 7, 7, 7],
    [-5, -1, -3, -2, -4]
]
for l in lists:
    print("Input :", l)
    print("Output:",
    sort_list(l)) print()
```

OUTPUT



2. Describe the Selection Sort algorithm's process of sorting an array. Selection Sort works by dividing the array into a sorted and an unsorted region. Initially, the sorted region is

empty, and the unsorted region contains all elements. The algorithm repeatedly selects the smallest element from the unsorted region and swaps it with the leftmost unsorted element, then moves the boundary of the sorted region one element to the right. Explain why Selection Sort is simple to understand and implement but is inefficient for large datasets. Provide examples to illustrate step-by-step how Selection Sort rearranges the

elements into ascending order, ensuring clarity in your explanation of the algorithm's mechanics and effectiveness.

Sorting a Random Array:

Input: [5, 2, 9, 1, 5, 6]

Output: [1, 2, 5, 5, 6, 9]

Sorting a Reverse Sorted Array:

Input: [10, 8, 6, 4, 2]

Output: [2, 4, 6, 8, 10]

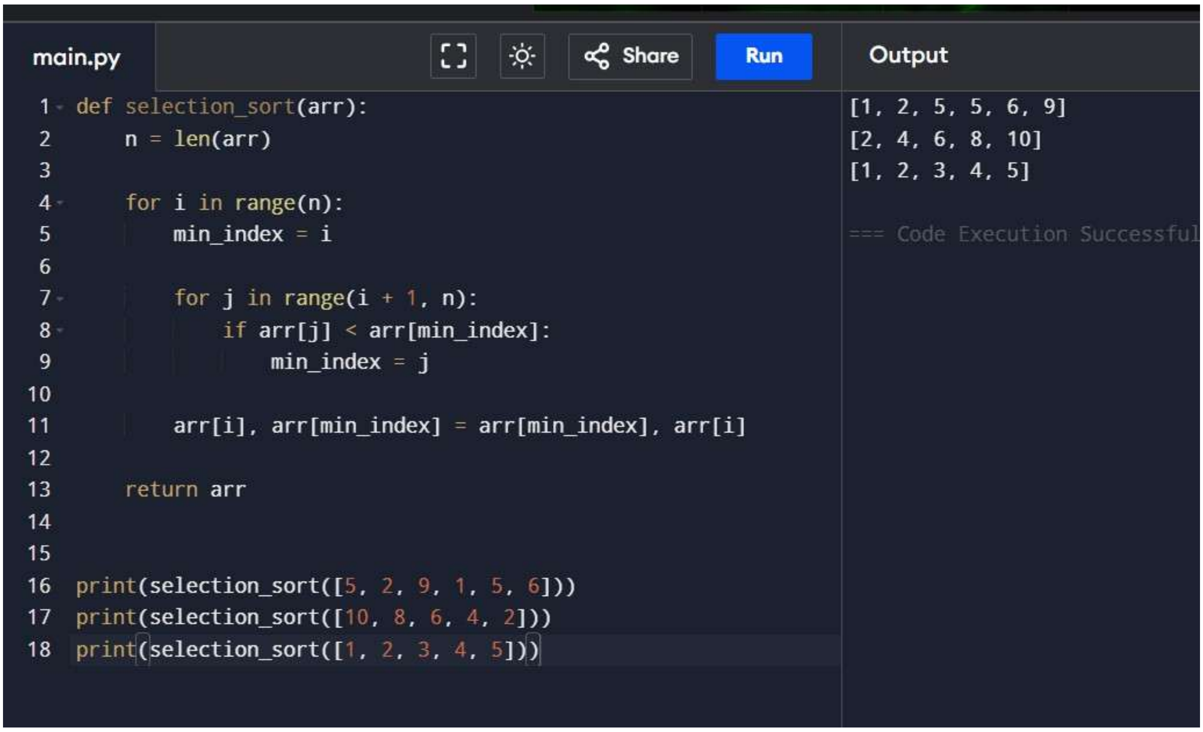
Sorting an Already Sorted

Array: Input: [1, 2, 3, 4, 5]

CODE

```
def
    selection_sort(arr):
    n = len(arr)
    for i in
        range(n):
        min_index =
        i
        for j in range(i + 1, n):
            if arr[j] < arr[min_index]:
                min_index = j
        arr[i], arr[min_index] = arr[min_index],
    arr[i] return arr
print(selection_sort([5, 2, 9, 1, 5, 6]))
print(selection_sort([10, 8, 6, 4, 2]))
print(selection_sort([1, 2, 3, 4, 5]))
```

OUTPUT



3. Write code to modify bubble_sort function to stop early if the list becomes sorted before all passes are completed.

CODE

```
def
   bubble_sort(arr):
   n = len(arr)

   for i in range(n):
      swapped = False
      for j in range(0, n - i -
         1): if arr[j] > arr[j +
         1]:
            arr[j], arr[j + 1] = arr[j + 1], arr[j]
            swapped = True
      if not swapped:
         break
   return arr
print(bubble_sort([64,25,12,22,11]
))
print(bubble_sort([29,10,14,37,13]
)) print(bubble_sort([3,5,2,1,4]))
print(bubble_sort([1,2,3,4,5]))
print(bubble_sort([5,4,3,2,1]))
```

OUTPUT

main.py

```python
def bubble_sort(arr):
    n = len(arr)

    for i in range(n):
        swapped = False

        for j in range(0, n - i - 1):

            if arr[j] > arr[j + 1]:
                arr[j], arr[j + 1] = arr[j + 1], arr[j]
                swapped = True

        if not swapped:
            break

    return arr


print(bubble_sort([64,25,12,22,11]))
print(bubble_sort([29,10,14,37,13]))
print(bubble_sort([3,5,2,1,4]))
print(bubble_sort([1,2,3,4,5]))
print(bubble_sort([5,4,3,2,1]))
```

Output

```
[11, 12, 22, 25, 64]
[10, 13, 14, 29, 37]
[1, 2, 3, 4, 5]
[1, 2, 3, 4, 5]
[1, 2, 3, 4, 5]

=== Code Execution Successful ===
```

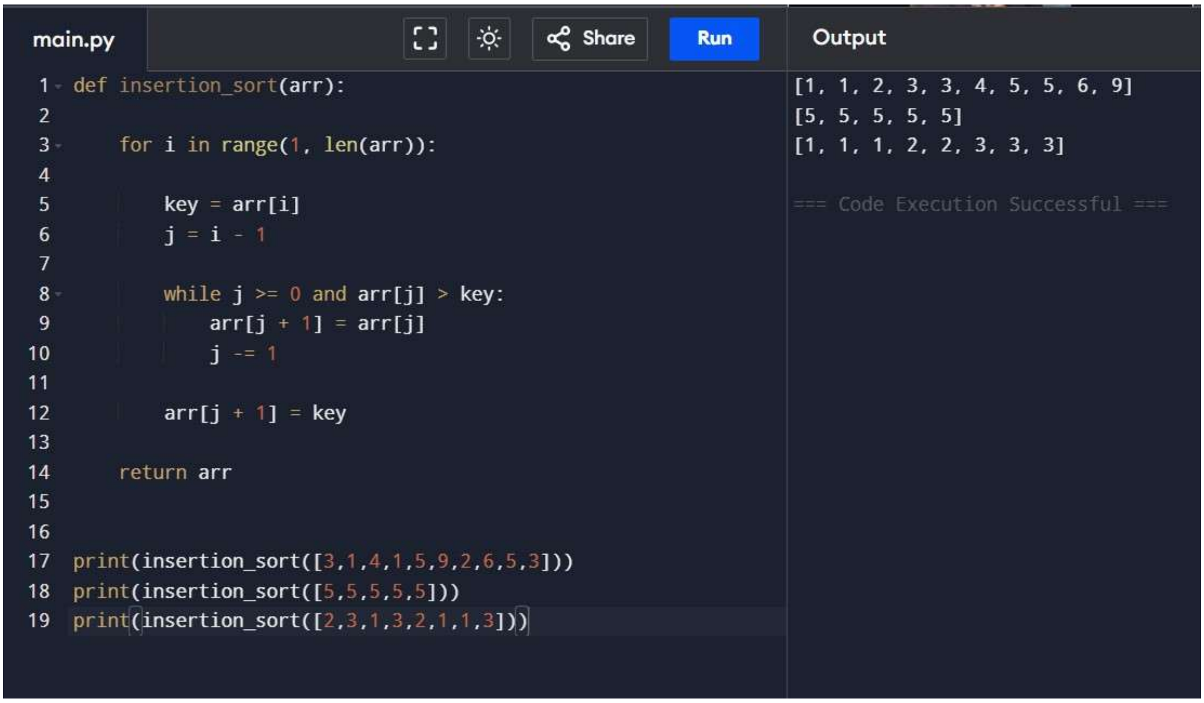
4. Write code for Insertion Sort that manages arrays with duplicate elements during the sorting process. Ensure the algorithm's behavior when encountering duplicate values, including whether it preserves the relative order of duplicates and how it

affects the overall sorting

outcome. CODE

```
def insertion_sort(arr):
    for i in range(1, len(arr)):
        key = arr[i]
        j = i - 1
        while j >= 0 and arr[j] > key:
            arr[j + 1] = arr[j]
            j -= 1
        arr[j + 1] = key
    return arr
print(insertion_sort([3,1,4,1,5,9,2,6,5,3])
) print(insertion_sort([5,5,5,5,5]))
print(insertion_sort([2,3,1,3,2,1,1,3]))
```

OUTPUT



5. Given an array arr of positive integers sorted in a strictly increasing order, and an integer k. return the kth positive integer that is missing from this array.

Example 1:

Input: arr = [2,3,4,7,11], k = 5

Output: 9

Explanation: The missing positive integers are [1,5,6,8,9,10,12,13,...]. The 5th missing positive integer is 9.

Example 2:

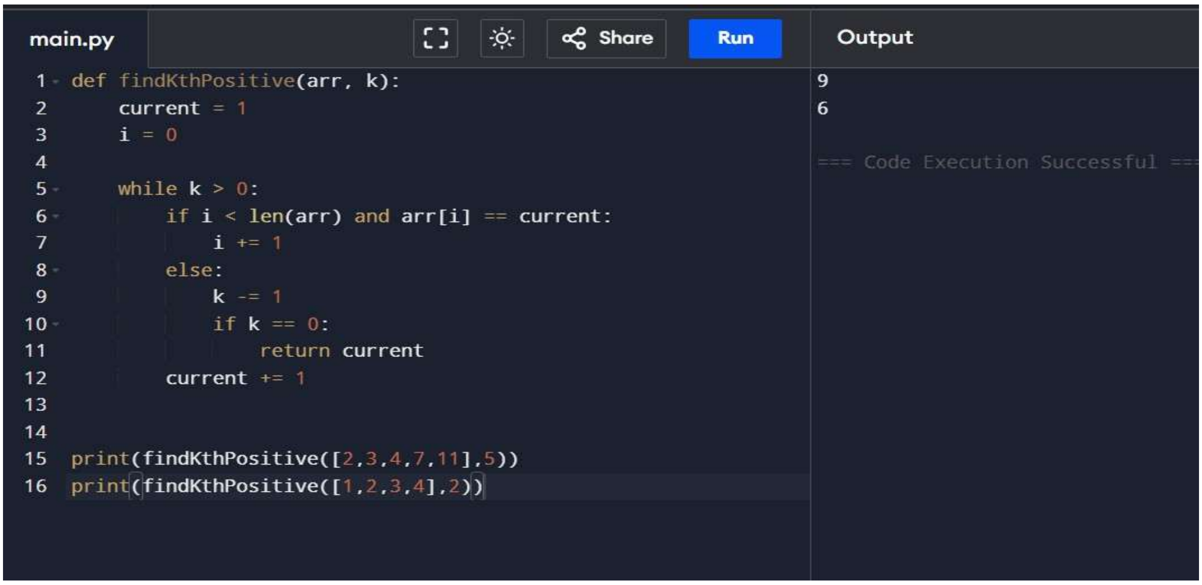
Input: arr = [1,2,3,4], k =

2 Output: 6

Explanation: The missing positive integers are [5,6,7,...]. The 2nd missing positive integer is 6.

CODE

```
def findKthPositive(arr, k):
    current = 1
    i = 0
    while k > 0:
        if i < len(arr) and arr[i] ==
            current: i += 1
        else:
            k -= 1
            if k == 0:
                return
        current current
        += 1
print(findKthPositive([2,3,4,7,11],
5))
print(findKthPositive([1,2,3,4],2))
```



OUTPUT

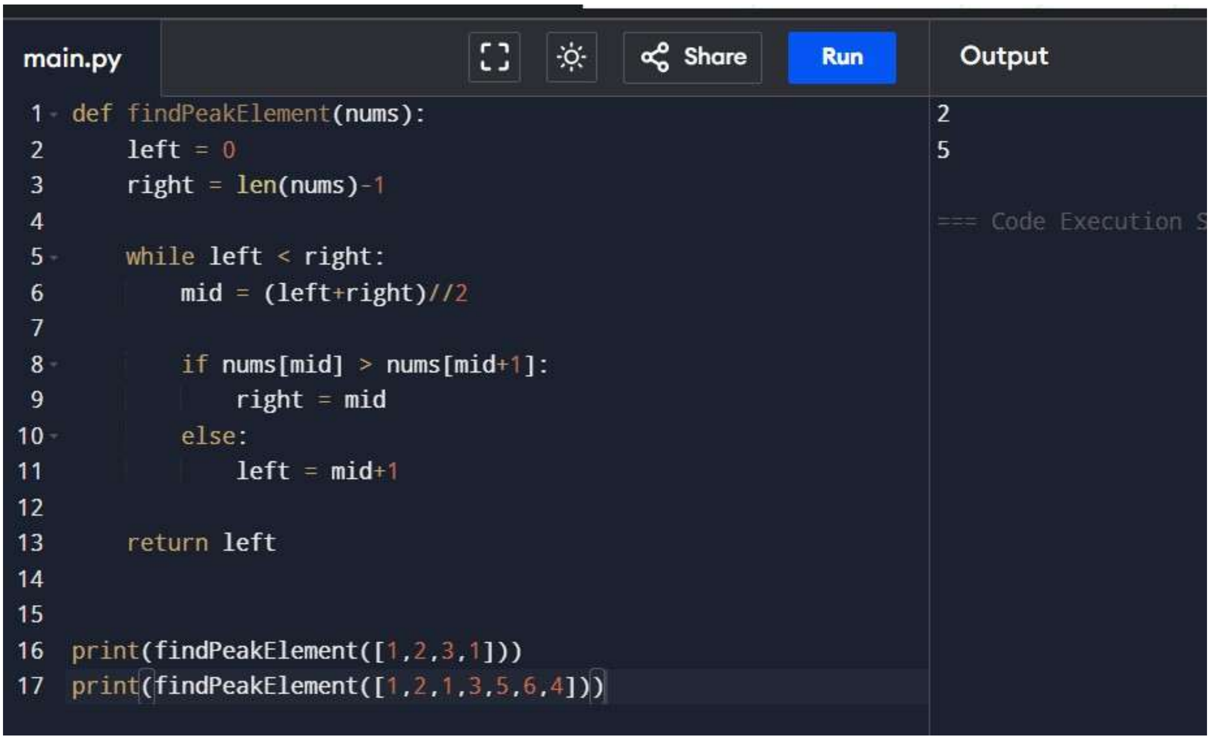
6. A peak element is an element that is strictly greater than its neighbors. Given a 0-indexed integer

array nums, find a peak element, and return its index. If the array contains multiple peaks, return the index to any of the peaks. You may imagine that nums[-1] = nums[n] = -∞. In other words, an element is always considered to be strictly greater than a neighbor that is outside the array. You must write an algorithm that runs in O(log n) time.

CODE

```
def
  findPeakElement(nums
  ): left = 0
  right = len(nums)-1
  while left < right:
    mid = (left+right)//2
    if nums[mid] >
      nums[mid+1]: right = mid
    else:
      left = mid+1
  return left
print(findPeakElement([1,2,3,1]))
print(findPeakElement([1,2,1,3,5,6,4]))
```

OUTPUT



7. Given two strings needle and haystack, return the index of the first occurrence of needle in haystack, or -1 if needle is not part of haystack.

CODE

def strStr(haystack, needle):

return

haystack.find(needle)

print(strStr("sadbutsad","sad"))

print(strStr("leetcode","leeto"))’

OUTPUT

8.Given an array of string words, return all strings in words that is a substring of another word. You can return the answer in any order. A substring is a contiguous sequence of characters within a string

CODE

```
def
    stringMatching(words
    ): ans=[]
    for i in range(len(words)):
        for j in
        range(len(words)):
            if i!=j and words[i] in words[j]:
                ans.append(words[i])
                brea
    k return ans
print(stringMatching(["mass","as","hero","superhero"]))
print(stringMatching(["leetcode","et","code"]))
print(stringMatching(["blue","green","bu"]))
```

OUTPUT

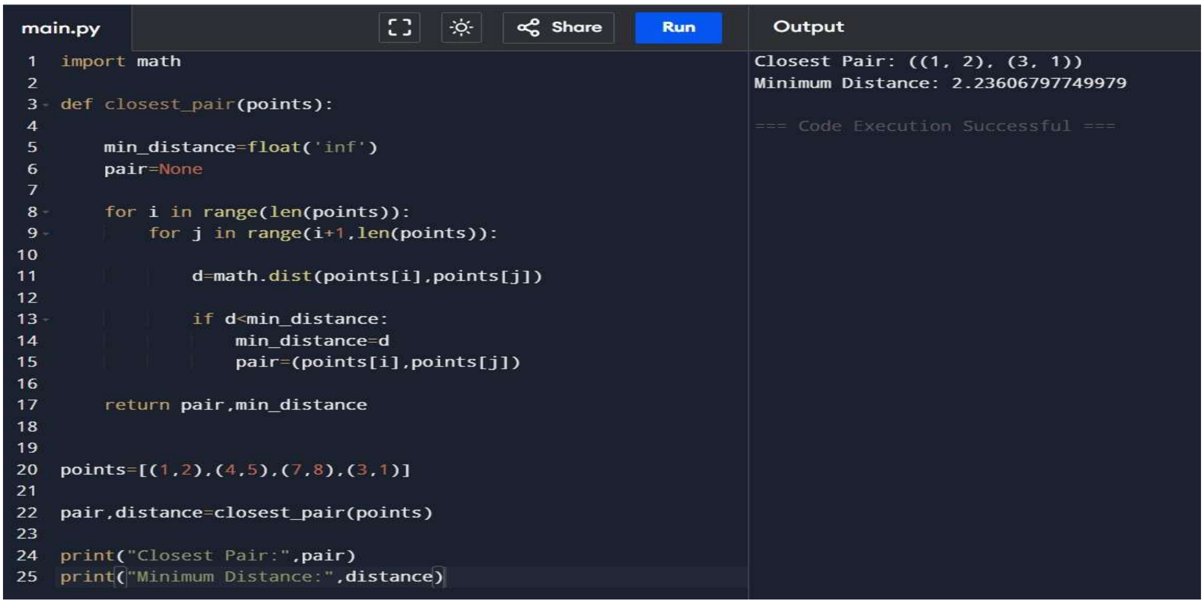
9. Write a program that finds the closest pair of points in a set of 2D points using the brute force approach.

CODE

```
import math


def closest_pair(points):
   min_distance=float('inf')
   pair=None
   for i in range(len(points)):
      for j in range(i+1,len(points)):
         d=math.dist(points[i],points[j
         ]) if d<min_distance:
            min_distance=d
            pair=(points[i],points[j])
   return pair,min_distance
points=[(1,2),(4,5),(7,8),(3,
1)]
pair,distance=closest_pair(points)
print("Closest Pair:",pair)
print("Minimum
Distance:",distance)
```

OUTPUT



10. **Write a program to find the closest pair of points in a given set using the brute force approach. Analyze the time complexity of your implementation. Define a function to calculate the Euclidean distance between two points. Implement a function to find the closest pair of points using the brute force method. Test your program with a sample set of points and verify the correctness of your results. Analyze the time complexity of your implementation. Write a brute-force algorithm to solve the convex hull problem for the following set S of points? P1 (10,0)P2 (11,5)P3 (5, 3)P4 (9, 3.5)P5 (15, 3)P6 (12.5, 7)P7 (6, 6.5)P8 (7.5, 4.5).How do you modify your brute force algorithm to handle multiple points that are lying on the sameline?**

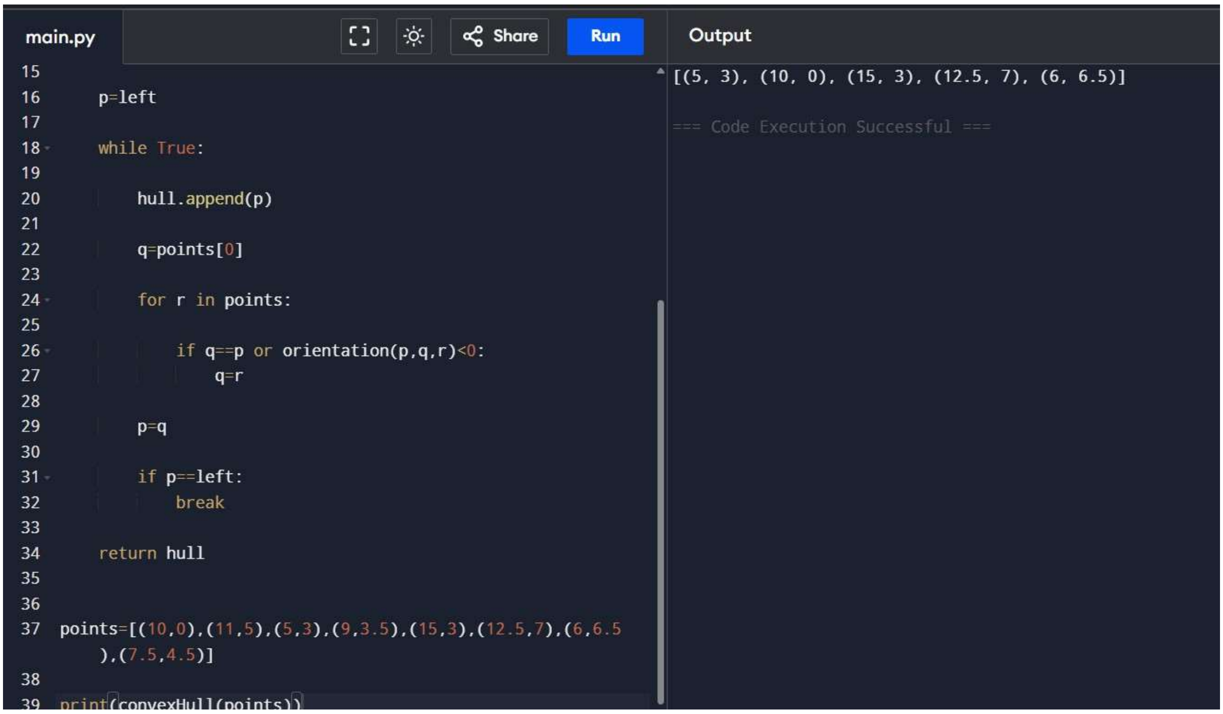
CODE

```
def orientation(p,q,r):
  return (q[0]-p[0])*(r[1]-p[1])-(q[1]-p[1])*(r[0]-
p[0]) def convexHull(points):
  n=len(points)
  if n<3:
    return points
  hull=[]
  left=min(points
  ) p=left
  while True:
    hull.append(p)
    q=points[0]

    for r in points:
      if q==p or orientation(p,q,r)<0:
        q=r
    p=q
    if
      p==lef
      t:
      break
```

```
    return hull
points=[(10,0),(11,5),(5,3),(9,3.5),(15,3),(12.5,7),(6,6.5),(7.5,4.5)]
print(convexHull(points))
```

OUTPUT



11. Write a program that finds the convex hull of a set of 2D points using the brute force approach.

CODE

```
def orientation(p,q,r):
    return (q[0]-p[0])*(r[1]-p[1])-(q[1]-p[1])*(r[0]-
p[0]) def convexHull(points):
    left=min(points
    ) hull=[]
    p=left
    while
    True:
        hull.append(
        p)
        q=points[0]
        for r in
        points:
            if q==p or orientation(p,q,r)<0:
                q=r
        p=q
        if
```

p==lef

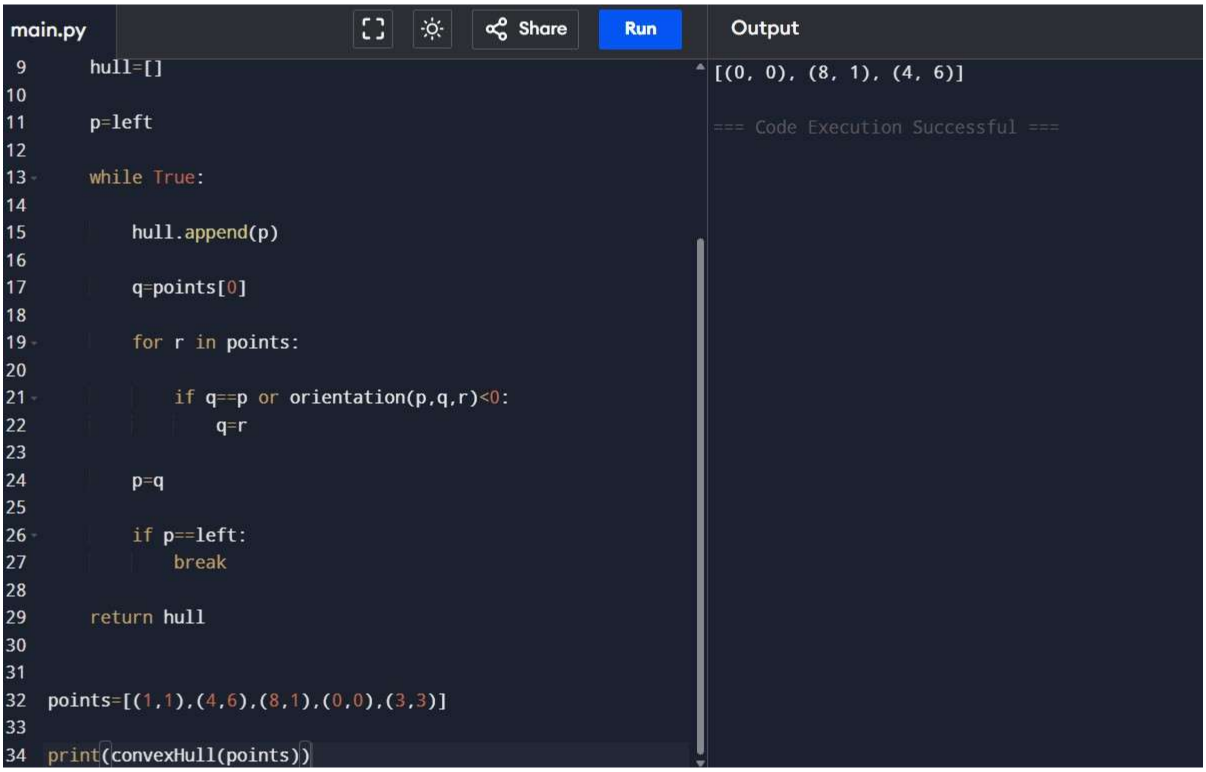
t:

break

```
    return hull
points=[(1,1),(4,6),(8,1),(0,0),(3,3)
]


print(convexHull(points))
```

OUTPUT



12. 1 You are given a list of cities represented by their coordinates. Develop a program that utilizes exhaustive search to solve the TSP. The program should:

1. Define a function distance(city1, city2) to calculate the distance between two cities (e.g., Euclidean distance).

2. Implement a function tsp(cities) that takes a list of cities as input and performs the following:

o Generate all possible permutations of the cities (excluding the starting city) using itertools.permutations.

o For each permutation (representing a potential route):

□ **Calculate the total distance traveled by iterating through the path and summing the distances between consecutive cities.**

□ **Keep track of the shortest distance encountered and the corresponding path.**

o **Return the minimum distance and the shortest path (including the starting city at the beginning and end).**

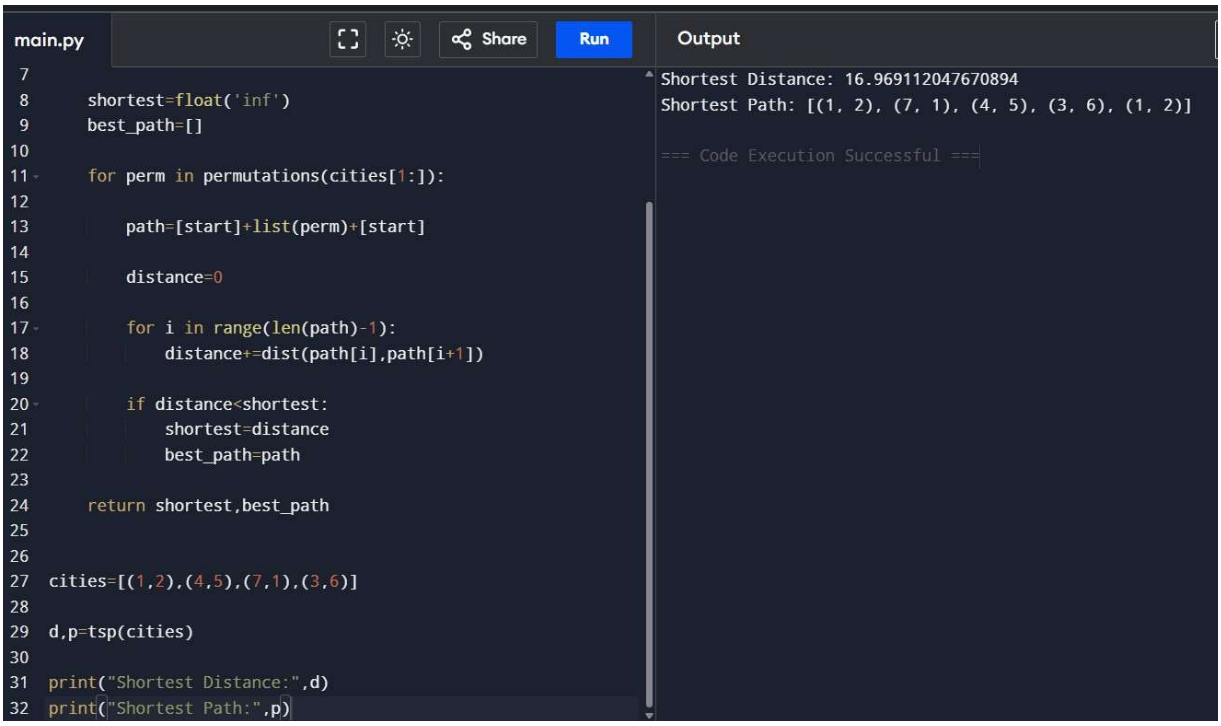
3. Include test cases with different city configurations to demonstrate the program's functionality. Print the shortest distance and the corresponding path for each test case.

CODE

```
from itertools import permutations
from math import dist
def tsp(cities):
  start=cities[0]
  shortest=float('inf')
  best_path=[]
  for perm in permutations(cities[1:]):
    path=[start]+list(perm)+[start]
    distance=0
    for i in range(len(path)-1):
      distance+=dist(path[i],path[i+1])
    if
      distance<shorte
      st:
      shortest=distanc
      e
      best_path=path
  return shortest,best_path
cities=[(1,2),(4,5),(7,1),(3,6
)]
d,p=tsp(cities)
print("Shortest
Distance:",d)
```

```
print("Shortest Path:",p)
```

OUTPUT



```
main.py

    shortest=float('inf')
    best_path=[]

    for perm in permutations(cities[1:]):

        path=[start]+list(perm)+[start]

        distance=0

        for i in range(len(path)-1):
            distance+=dist(path[i],path[i+1])

        if distance<shortest:
            shortest=distance
            best_path=path

    return shortest,best_path


cities=[(1,2),(4,5),(7,1),(3,6)]

d,p=tsp(cities)

print("Shortest Distance:",d)
print("Shortest Path:",p)
```

```
Output
Shortest Distance: 16.969112047670894
Shortest Path: [(1, 2), (7, 1), (4, 5), (3, 6), (1, 2)]

=== Code Execution Successful ===
```

13. You are given a cost matrix where each element cost[i][j] represents the cost of assigning worker i to task j. Develop a program that utilizes exhaustive search to solve the assignment problem. The program should Define a function total_cost(assignment, cost_matrix) that takes an assignment (list representing worker-task pairings) and the cost matrix as input. It iterates through the assignment and calculates the total cost by summing the corresponding costs from the cost matrix Implement a function assignment_problem(cost_matrix) that takes the cost matrix as input and performs the following Generate all possible permutations of worker indices (excluding repetitions). CODE

```
from itertools import
permutations def
assignment_problem(cost):
    n=len(cost)
    minimum=float('in
    f') best=None
    for perm in permutations(range(n)):
        total=0
        for worker in range(n):
            total+=cost[worker][perm[worker]]
```

if

total<minimu

m:

minimum=tot

al best=perm

return

best,minimum

cost=[[3,10,7],

[8,5,12],

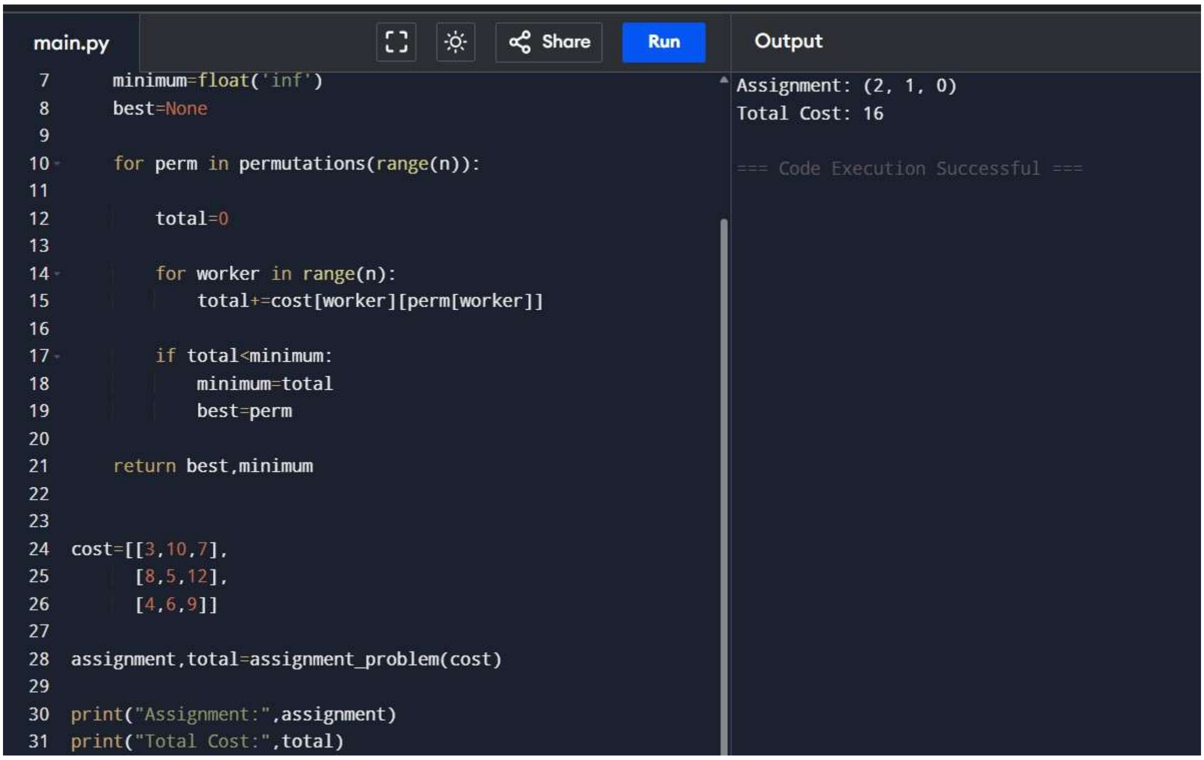
[4,6,9]]

assignment,total=assignment_problem(c

ost) print("Assignment:",assignment)

print("Total Cost:",total)

OUTPUT



14. You are given a list of items with their weights and values. Develop a program that utilizes exhaustive search to solve the 0-1 Knapsack Problem. The program should:

1. Define a function total_value(items, values) that takes a list of selected items (represented by their indices) and the value list as input. It iterates through the selected items and calculates the total value by summing the corresponding values from the value list.

2. Define a function is_feasible(items, weights, capacity) that takes a list of selected items (represented by their indices), the weight list, and the knapsack capacity as input. It checks if the total weight of the selected items exceeds the capacity. CODE

main.py

```
from itertools import permutations

def assignment_problem(cost):

    n=len(cost)

    minimum=float('inf')
    best=None

    for perm in permutations(range(n)):

        total=0

        for worker in range(n):
            total+=cost[worker][perm[worker]]

        if total<minimum:
            minimum=total
            best=perm

    return best,minimum


cost=[[3,10,7],
      [8,5,12],
      [4,6,9]]
```

Output

```
Assignment: (2, 1, 0)
Total Cost: 16

=== Code Execution Successful ===
```